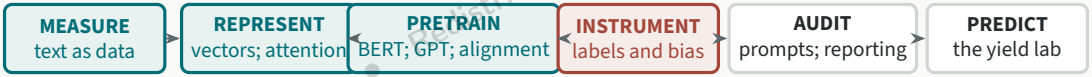
Session 5.4

The LLM as a Measurement Instrument

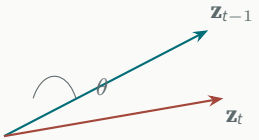
What it can label, what the labels are worth, and how to correct what they get wrong



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Cosine Similarity as an Economic Variable



$\text{sim} = \cos\theta$: same direction 1, orthogonal 0, opposed -1

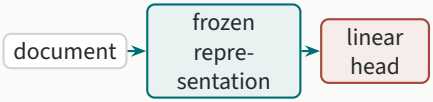
$\delta_t = 1 - \text{sim}(\mathbf{z}_{t-1}, \mathbf{z}_t)$
how far this statement moved

- $\text{sim}(i,j) = \mathbf{z}_i^\top \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$ is invariant to length, and length tracks firm size.
- **Cosine distance** over consecutive statements is continuous where a hawk–dove label is binary.
- To call δ_t a shock, clear two layers: what markets priced, and whether the bank merely responded to conditions.
- **The geometry gives a number. Whether it is a shock is a claim about the design.**

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Source: Miao & Feng, *AI for Economists*, ch. 13, "语义相似度计算"; Acosta & Meade (2015).

Document Classification, With Labels and Without



- **Supervised**: freeze the representation, fit a linear head on a few hundred labels. The labels are the binding constraint.
- **Zero-shot**: embed a class description with the document; take the nearest class.
- Raw encoder vectors are anisotropic: sentences sharing frequent words look close.
- **Needing no labels is paid for in validation: the class wording moves the result.**

Source: Miao & Feng, *AI for Economists*, ch. 13; Reimers & Gurevych (2019).

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The Standard These Measures Are Held To

measure	study	text operation	how it was validated
corporate culture	Li, Mai, Shen & Yan (2021)	a dictionary grown from seed words	later performance; enforcement actions
macro nowcast	Bybee, Kelly & Manela (2023)	topic shares as predictors	added forecast power for output
political risk	Hassan & Hollander (2019)	counts of political word pairs	lobbying spending
communication change	Acosta & Meade (2015)	cosine distance between statements	policy turning points; descriptive

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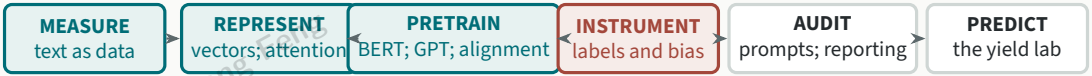
- **The last column is what makes any of them a measure — and every row still yields a predictive proxy, not an identified effect.**

Source: Miao & Feng, *AI for Economists*, ch. 13, table of text measures in economics.

Session 5.4

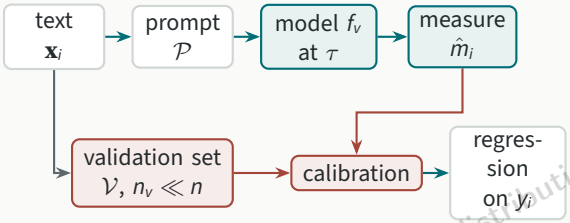
The Instrument

A prompt, a model version and a temperature: where a text measure comes from now



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The Measurement Chain, End to End



- Every box is a decision that has to be reported: corpus, prompt text, model version, temperature, and how $\mathcal{V}$ was drawn.
- $\mathcal{V}$ is the only link that puts $\hat{m}_i$ back on the scale of the construct.
- **Everything to the right of the measure is econometrics, not language modelling.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, figure of the measurement chain.

Fine-Tune an Encoder, or Prompt a Model

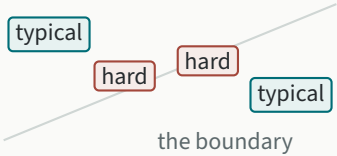
criterion	fine-tuned encoder	prompted model
labels	hundreds in hand, or affordable	none, or costlier than the answer
task definition	settled; the manual is written	still being revised
sample size	hundreds of thousands	thousands to tens of thousands
reproducibility	frozen weights, reruns to the digit	version disclosed, robustness re-checked
what it leans on	dense domain vocabulary	common-sense and pragmatic inference

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- **Cost structure decides this, not accuracy: fine-tuning owns an instrument, prompting rents one. Or prompt a few hundred cases to settle the definitions, then fine-tune on them.**

Source: Miao & Feng, *AI for Economists*, ch. 14, table comparing encoder fine-tuning with prompting.

Zero- and Few-Shot Labelling, and Which Examples to Show

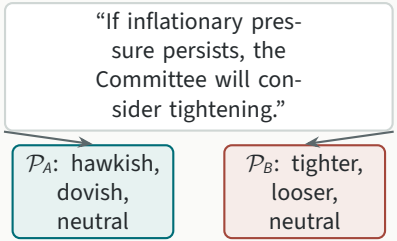


- One equation for the whole section: $\hat{m}_i = \arg\max_c \, p(c \mid \mathbf{x}_i, \mathcal{P}; \boldsymbol{\theta})$, with $\boldsymbol{\theta}$ fixed and the prompt doing every bit of task adaptation.
- **Zero-shot** removes the fixed cost of labelling; **few-shot** buys calibration with a few demonstrations.
- **Pick demonstrations at the boundary, not at the centre**: consistency across a sample is decided by the hard cases.

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Source: Miao & Feng, *AI for Economists*, ch. 14, "零样本与少样本标注"; Gilardi et al. (2023).

One Statement, Two Wordings



the drift is systematic: inter-coder reliability cannot see it

claude-sonnet-5, Claude Code 2.1.278 · 2026-09-20 · prompts 846fe144, 3307f8f3, a47b80db

- The sentence is a conditional: nothing is announced.
- $\mathcal{P}_A$ folds "hawkish" into "announces tightening"; $\mathcal{P}_B$ leaves room for a leaning. On an unconditional sentence the two agree.
- This session's own stance run, 4 prompt variants over 48 statements: κ 0.746 reworded, 0.843 reordered, 0.717 on an identical repeat.
- **Write conditionals into the class definition and test on boundary cases.**

Source: Miao & Feng, *AI for Economists*, ch. 14, example box on prompt perturbation.

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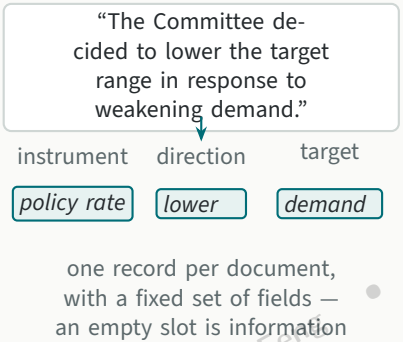
Sentiment: Where the Advantage Actually Comes From

- A dictionary reads a document as an unordered set of words: negation, conditionals and irony sit outside the model rather than being handled badly by it.
- **Reach**. Nineteenth-century newspapers are dense with scanning error, where keyword matching is brittle and a semantic judgement is not. The gain is which archives become usable at all.
- **Excess predictive power**. Daily firm news scored by a model and sorted into a long–short portfolio keeps a positive return spread after costs — and tell the model to judge words literally instead, and that power disappears.
- **Structured sentiment**. Split macro news into drivers — inflation worry, supply-chain strain, weak demand — and the nowcast improves where one positive–negative score does not.
- **The advantage is context, which is also where the error is hardest to diagnose from word counts.**

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Source: Loughran & McDonald (2011); Lopez-Lira & Tang (2023); Hartley (2025); BIS (2025).

Structured Extraction: Fields, Not Labels



- The output is a record, not a class: acquirer, target, value, date.
- Prompted extraction beats few-shot baselines on spans, tokens and relations in clinical text — jargon-dense and inconsistently named, as economic text is.
- A narrow templated task still favours a small fine-tuned model.
- **Session 6 turns this record into a typed tool call, where the schema is enforced.**

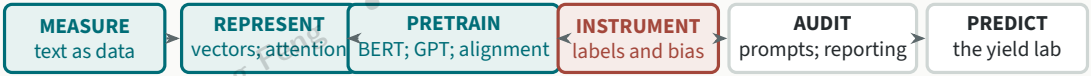
Source: Miao & Feng, *AI for Economists*, ch. 14, "结构化抽取"; Agrawal et al. (2022).

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Session 5.4

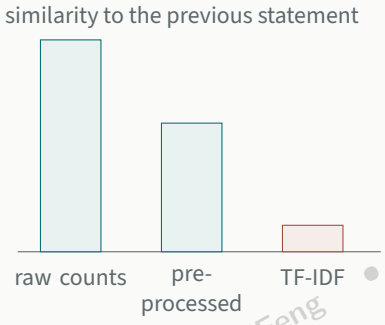
Measures That Were Estimated

Similarity, topic shares and labels are all output of a first stage



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The Same Two Statements Under Three Weightings

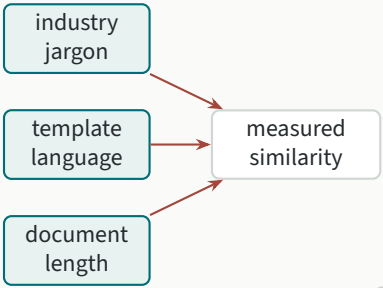


- The December 2008 statement — the meeting that took the funds-rate target to its floor — reads as a light edit on raw counts and as a near-complete rewrite under TF-IDF.
- Raw counts are dominated by shared template text; TF-IDF gives its weight to what only this statement says.
- **Deck 5.1's pre-processing choices do not wash out downstream: they set the size of the variable you are about to regress.**

Source: Acosta & Meade (2015), consecutive statements under three weightings; Miao & Feng, ch. 13.

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Three Confounds Inside a Similarity



all three push it up, none of them is strategy

- **Industry**: one industry shares a vocabulary, so similarity is high before any strategy is compared.
- **Template**: mandated disclosure language repeats verbatim, with nothing behind it.
- **Length**: a longer document has a steadier average vector and looks closer to everything.
- **Deck 2.3's leakage frame is the template confound**: template sentences scored 0.97 on a random split and 0.50 — chance — with phrases held out.

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Source: Miao & Feng, *AI for Economists*, ch. 13, "验证挑战：三类混淆"; Session 2 lab notebook, appendix.

What to Do About Them

permutation
shuffle labels inside an industry

is it above the baseline?

residual
subtract the industry mean

remove the shared part

external
tie it to an observable outcome

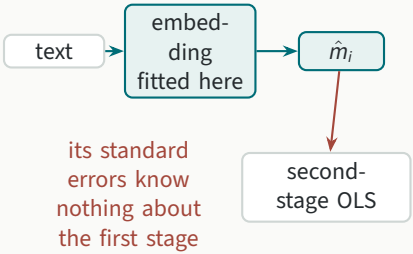
is it a measure of anything?

- **Permutation test**: shuffle firm labels within an industry, recompute the similarity distribution, and ask whether the observed value clears that baseline.
- **Residual embedding**: subtract the industry mean vector before measuring.
- **External validation**: relate the similarity to merger incidence or joint research spending.
- **The first two remove a confound. Only the third says the measure is of anything.**

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Source: Miao & Feng, *AI for Economists*, ch. 13; Gentzkow, Kelly & Taddy (2019).

The Generated-Regressor Problem

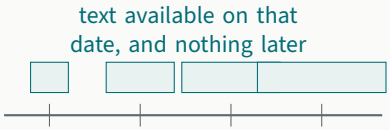


- A similarity, a topic share, a model score: each is an estimate carrying error from the model and the pooling rule.
- Treat it as observed and the second-stage standard errors are too small.
- Worst where the first stage is small: an embedding fitted on a few hundred statements.
- **Resample documents, re-embed, re-estimate, and report that interval.**

Source: Miao & Feng, *AI for Economists*, ch. 13, "生成回归量问题".

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Re-Estimating at Each Release Is a Design, Not a Detail

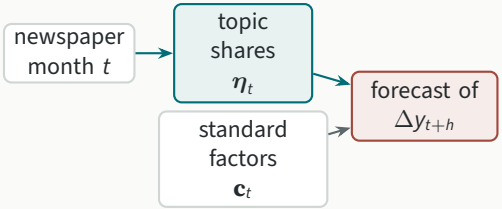


- The euro-area exercise refits its word vectors at every press conference, from the text available that day, and forecasts core inflation from them.
- Fitting it once on the full sample leaks the future into the representation, not just the regression.
- Session 2's leakage rule reaches the first stage: the vectors belong to the training window.
- **It costs compute and buys a claim the forecast could have been made.**

Source: Araujo, Bokan, Comazzi & Lenza, real-time word vectors for inflation forecasting.

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Nowcasting: Text Factors Enter Like Any Other Factor



$\boldsymbol{\eta}_t$ came out of a first-stage fit: the interval has to be bootstrapped through it

- Quarterly output is published about a quarter late; a nowcast has to speak before the release.
- Monthly topic shares enter the usual predictive regression beside surveys and lagged output:

$$\Delta y_{t+h} = \alpha + \boldsymbol{\beta}^\top \boldsymbol{\eta}_t + \boldsymbol{\gamma}^\top \mathbf{c}_t + \varepsilon_{t+h}.$$

- **Text still adds forecast power with the standard factors controlled — but bootstrap text, topics and regression together.**

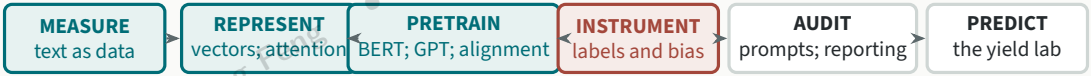
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Source: Bybee, Kelly & Manela (2023); Miao & Feng, *AI for Economists*, ch. 13.

Session 5.4

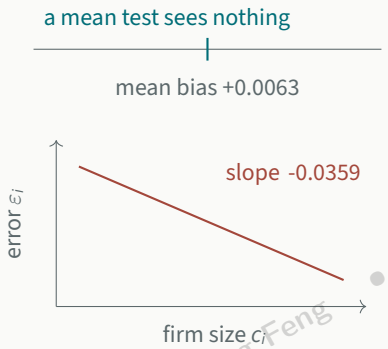
The Error, and What to Do About It

A validation set, a calibration equation, and a budget for hand coding



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The Error Is Not the Classical Kind



- $\hat{m}_i = m_i^* + \varepsilon_i$, and ε_i moves with m_i^* (models avoid extreme judgements) and with the covariates.
- On the book's simulated reports the measure correlates 0.851 with the truth at mean bias +0.0063: a mean test passes it.
- The error on firm size has slope -0.0359, standard error 0.0019: large firms are understated.
- **What matters is the covariance of the error with the regressors, not its mean.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "测量误差的结构"; Ludwig & Mullainathan (2025).

On the Right-Hand Side, or on the Left

$$y_i = \alpha + \beta \hat{m}_i + \boldsymbol{\gamma}^\top \mathbf{c}_i + u_i$$

biased; the bias leaks into γ as well

$$\hat{m}_i = \alpha + \boldsymbol{\gamma}^\top \mathbf{c}_i + u_i$$

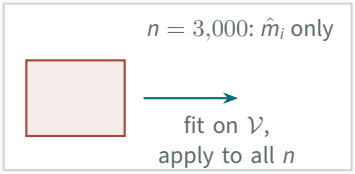
consistent if $\mathbb{E}[\varepsilon_i \mid \mathbf{c}_i] = 0$;
only precision is lost

- As a **regressor**, $\hat{m}_i$ brings attenuation plus a non-classical part whose direction cannot be signed in advance.
- As an **outcome** — what makes a firm write this way? — least squares stays consistent as long as the error has conditional mean zero.
- **Opposite diagnostics: on the right, test the error against the regressors; on the left, test its conditional mean. Carrying one check across is the common mistake.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "测量误差的结构".

Correcting It With a Human Validation Set



$\mathcal{V}$: $n_v = 200$,
both m_i^* and $\hat{m}_i$

- Hand-code a validation set $\mathcal{V}$ of size $n_v \ll n$ carrying both the human m_i^* and the model's $\hat{m}_i$.
- Step one, on $\mathcal{V}$: $\hat{m}_i^* = \hat{a}_0 + \hat{a}_1 \hat{m}_i + \hat{\mathbf{a}}_2^\top \mathbf{c}_i$. Step two, on all n: regress y_i on $\hat{m}_i^*$ and $\mathbf{c}_i$.
- $\mathbf{c}_i$ has to be in the calibration equation — it is what absorbs the systematic link between the error and the covariates.
- **Do not use the second step's textbook standard errors: the calibration is itself estimated, so bootstrap both samples.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "用人工验证集做偏误修正".

What the Correction Assumes

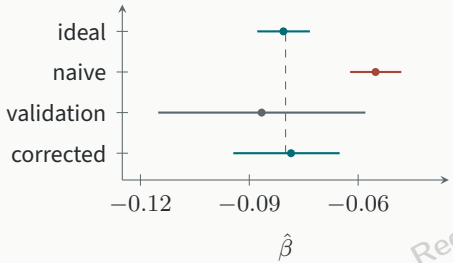
random draw $\mathcal{V}$ from the full sample	coding the easy ones first breaks it
extrapolation the fit holds outside $\mathcal{V}$	rare wording may not be in $\mathcal{V}$
exclusion the residual of $\hat{m}$ is inert	what is left must not reach y_i

- **One**: $\mathcal{V}$ is a random draw. Hand-coding whatever is quickest to read breaks it.
- **Two**: the calibration extrapolates beyond $\mathcal{V}$. If the model fails only on unusual wording, $\mathcal{V}$ may contain none of it.
- **Three**: given $\hat{m}_i$ and $\mathbf{c}_i$, the relation between m_i^* and y_i no longer depends on what is left in $\hat{m}_i$.
- **State all three. Each is a claim about how the labels were made, which only the researcher can answer.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, the three assumptions of the two-step estimator.

Four Estimators, and the Slope That Explains Them



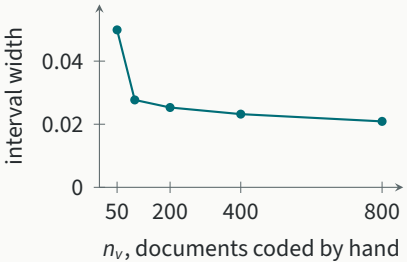
dashed: the true $\beta = -0.080$

- Ideal, on m_i^*: -0.0806. Naive, on $\hat{m}_i$: -0.0552, 31.0 per cent from the truth and shrunk toward zero.
- Calibration on $\mathcal{V}$: $0.1423 + 0.7031\,\hat{m} + 0.0048\,c$, and that slope below one *is* the attenuation.
- Validation set alone -0.0866 at width 0.0570; corrected -0.0785, 1.9 per cent out, at 0.0293 — 49 per cent narrower.
- **200 hand-coded documents buy back full-sample precision.**

Source: Miao & Feng, *AI for Economists*, ch. 14, worked example on simulated annual reports.

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How Many Documents to Hand-Code

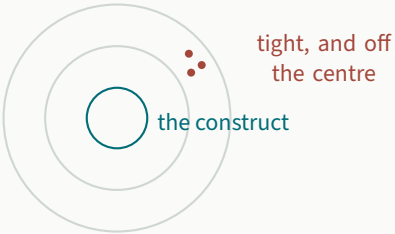


- Width falls in n_v at a decreasing rate: 49 per cent narrower from 50 to 200 labels, a further 17 per cent from 200 to 800.
- At small n_v the calibration's own error dominates; past a few hundred the full-sample residual variance binds.
- Hold the two 49 per cents apart: the last page compared two *estimators* at $n_v = 200$; this compares two *validation sizes* for one estimator.
- **Buy enough labels to kill the bias, then stop enlarging the set.**

Source: Miao & Feng, *AI for Economists*, ch. 14, design calculation for the validation set.

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Consistency Is Not Validity



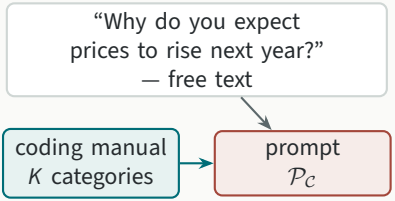
κ measures the spread of the darts, never the distance to the middle

- κ compares two coding processes; it is silent on whether either measures the construct.
- A model can be perfectly consistent about high and low commitment and still be unrelated to green investment.
- If the manual defines the construct badly, both coders agree on the wrong definition.
- **Validity is argued from outside: later emissions, capital expenditure, compliance.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, on reliability and construct validity.

Coding Open Survey Answers



- Narrative heterogeneity is the interest: do different attributions save and spend differently?
- Categories come from the manual *before* the data, or they become part of the result.
- The prompt must reproduce the operational definitions and the exclusion clauses.
- **Four rules: temperature zero; a real "cannot be coded" category; reasons before the label; the prompt archived.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, "开放式问题的自动编码"; Haaland et al. (2025).

Report κ by Class, Never Overall Alone

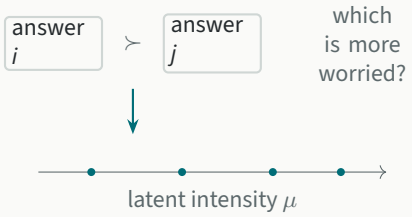
category	share	recall	κ	at the 0.75 threshold
supply-chain narrative	0.413	0.968	0.891	passes
monetary narrative	0.387	0.974	0.876	passes
opportunistic-pricing narrative	0.120	0.528	0.634	fails; act
cannot be coded	0.080	0.750	0.825	passes
overall	1.000	0.900	0.844	reads as excellent

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- A model is low-variance and possibly biased, so a repeat call adds nothing: audit 10 per cent of the 3,000 answers blind against a human gold standard.
- **Half of the low-frequency narrative is misfiled and the overall row hides it. Merging it lifts κ to 0.881 — by abolishing the distinction.**

Source: Miao & Feng, *AI for Economists*, ch. 14, per-class audit on simulated survey answers.

Intensity by Pairwise Comparison



the origin is not identified: only differences mean anything

- When the target is intensity, a direct score out of a hundred has no common scale; a pairwise judgement does.
- $\Pr(\mathbf{x}_i \succ \mathbf{x}_j) = e^{\mu_i}/(e^{\mu_i} + e^{\mu_j})$: Bradley–Terry, with μ by maximum likelihood and a randomised block design.
- Chat assistants are tuned with this model on human preferences between replies (Deck 5.3); here a model ranks human answers.
- **μ is identified only up to a shift: standardise it or use the ranks.**

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Source: Bradley & Terry (1952); Miao & Feng, *AI for Economists*, ch. 14, “连续强度的编码”.

The Validation Ladder

- **Pre-alignment**: hand-code one to two hundred answers and iterate the prompt against them — then retire them. They tuned the instrument, so they cannot grade it.
- **Gold-standard audit**: once the full sample is coded, blind-recode at least 10 per cent of it and compute Cohen's κ against the human labels.
- **Per-class report**: κ and the confusion matrix category by category. Below the threshold the fallback is hand coding or a coarser category, and the cost falls on the research question rather than on the statistic.
- One κ, three adjectives: the audit's 0.844 is excellent on the bands of Fleiss (1981), almost perfect on the 0.81–1.00 band of Landis & Koch (1977), and strong on McHugh (2012). The 0.75 threshold is Fleiss's; Landis & Koch has no such cut.
- **Name the scheme whenever a κ takes an adjective, and finish the ladder: the last rung is external validity, which none of the rungs below can supply.**

Source: Fleiss (1981); Landis & Koch (1977); McHugh (2012); Miao & Feng, *AI for Economists*, ch. 14.

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